MIL-W-81381/11C

5 October 1977

SUPERSEDING

MIL-W-81381/11B

15 November 1972

MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROCARBON/POLYIMIDE INSULATED, MEDIUM WEIGHT, SILVER COATED COPPER CONDUCTOR, 600 VOLTS, 200°C, NOMINAL 8.4 OR 15.4 MIL WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-81381.

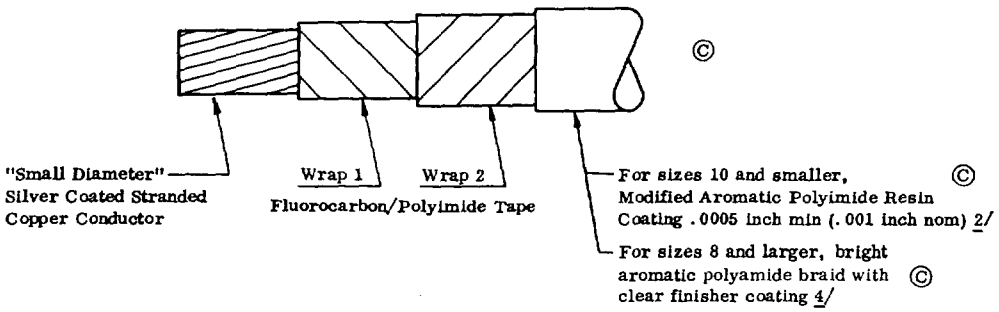


TABLE I. Construction details.

Part Number 1/	Wire Size	Conductor			Finished Wire				Insulation Tapes			
		Stranding (Number of Strands X AWG gage of strands)	Diameter (inches)		Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches) (min-max)	Weight (lbs/1000 ft)		Wrap 1		Wrap 2	
			(min)	(max)			(nom) 2/	(max)	Tape Code 3/	Overlap (%) (min)	Tape Code 3/	Overlap (%) (min)
M81381/11-24-*	24	19 x 36	.023	.024	24.3	.040-.044	2.0	2.1	0/2/.5	50	.1/1/.1	50
M81381/11-22-*	22	19 x 34	.029	.030	15.1	.045-.049	2.9	3.0				
M81381/11-20-*	20	19 x 32	.037	.038	9.19	.053-.057	4.4	4.6				
M81381/11-18-*	18	19 x 30	.046	.048	5.79	.063-.067	6.7	7.0				
M81381/11-16-*	16	19 x 29	.052	.054	4.52	.068-.073	8.3	8.7				
© M81381/11-14-*	14	19 x 27	.065	.068	2.88	.081-.086	12.9	13.2				
© M81381/11-12-*	12	37 x 28	.084	.087	1.90	.100-.105	19.2	20.2				
© M81381/11-10-*	10	37 x 26	.106	.110	1.19	.122-.127	29.5	31.1				
© M81381/11-8-*	8	133 x 29	.158	.166	.658	.188-.206	58.7	61.8			.5/1/.5	50
© M81381/11-6-*	6	133 x 27	.198	.208	.418	.228-.251	90.4	95.1				
© M81381/11-4-*	4	133 x 25	.250	.263	.264	.280-.306	141	148				
© M81381/11-2-*	2	665 x 30	.320	.340	.170	.350-.378	224	235				

1/ Part Number: The asterisks in the part number column, Tables I and II, shall be replaced by color code designators in accordance with MIL-STD-681, except that opaque dark yellow as defined in MIL-W-81381 shall be designated by the letter "N" and unpigmented polyimide resin coating shall be designated by the letter "C". Examples: Size 20, opaque dark yellow - M81381/11-20-N; same with orange stripe - M81381/11-20-N3.

2/ Nominal values are for information only. Nominal values are not requirements.

3/ Tape Codes: 0/2/.5 2 mil polyimide film/0.5 mil FEP fluorocarbon resin
.1/1/.1 0.1 mil FEP fluorocarbon resin/1 mil polyimide film/0.1 mil FEP fluorocarbon resin
.5/1/.5 0.5 mil FEP fluorocarbon resin/1 mil polyimide film/0.5 mil FEP fluorocarbon resin

FEP = Fluorinated Ethylene Propylene

© 4/ Braid (Sizes 8 and larger): Bright aromatic polyamide yarn, 200 denier, 100 filaments, tightly formed, uniform in appearance, treated with a clear finisher coating. The finisher coating shall be compatible with the 200°C temperature rating and the performance requirements of the insulated wire.

© denotes changes.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:21Z
Check the source to verify that this is the current version before use.

TABLE II. Performance details.

Part Number	Durability Test Load for		Abrasion Resistance				Bend Testing					
							Mandrel diameter (inches) (±3%)				Test Load (lbs) (±3%)	
	Color Markings (grams)	Insulation Coatings (lbs)	Weight Support Bracket	Weight (lbs)	Tension Load (lbs)	Resistance (inches of tape) (min)	Life Cycle (Oven & bend tests) 2/	Cold Bend Test	Wrap Test	Wrinkle Test ©	Life Cycle (Oven & bend tests) 2/	Cold Bend Test
M81381/11-24-*	75	.75	A	1.0	1.0	12	.250	.250	.125	.156	.500	
M81381/11-22-*	100	1.00	A	1.0	1.0	14	.250	.250	.125	.188	.750	
M81381/11-20-*	100	1.00	A	1.0	1.0	16	.250	.250	.125	.250	.750	
M81381/11-18-*	150	1.00	A	1.0	1.0	16	.375	.375	.250	.312	1.00	
M81381/11-16-*	150	1.00	A	1.0	1.0	20	.375	.375	.250	.375	1.00	
M81381/11-14-*	150	1.00	B	1.0	2.0	24	.500	.375	.375	.500	2.00	
M81381/11-12-*	150	1.00	B	1.0	2.0	26	.750	.750	.375	.750	2.00	
M81381/11-10-*	150	1.00	B	1.0	2.0	26	.750	.750	.375	1.00	3.00	
M81381/11-8-*	150	1/	©C	4.25	2.0	18	2.00	2.00	1.00	1/	3.00	
M81381/11-6-*	150	1/	C	4.25	2.0	27	4.50	4.50	2.00	1/	3.00	
M81381/11-4-*	150	1/	C	4.25	2.0	27	6.00	6.00	3.00	1/	4.00	
M81381/11-2-*	150	1/	C	4.25	2.0	27	6.00	6.00	4.00	1/	6.00	

1/ Test not applicable

2/ Also for bend tests after immersion

WIRE RATINGS AND ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392° F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
BLOCKING: Oven temperature, 200 ±2° C (392 ±3.6° F)
© BRAID FRAYING: The aromatic polyamide braid with finisher (wire sizes 8 and larger) shall be non-fraying and shall be firmly adherent to the underlying insulation. This examination shall be included in the MIL-W-81381 quality conformance inspection as a Group II characteristic, one specimen to be examined from each sample unit.
© COLOR: Sizes 24 through 10: As specified in contract or order in accordance with MIL-W-81381.
Sizes 8 through 2: The natural color of the finisher-treated aromatic polyamide braid (off-white to amber) is preferred (color designator "C"). Other colors, varying with the supplier, are available. The MIL-STD-104 color limits and the extended color limits of MIL-W-81381 are not applicable to these sizes.
FLAMMABILITY: 3 sec (max) after-flame
3.0 inches (max) flame travel
No flaming of tissue paper
HUMIDITY RESISTANCE: 5 megohms-1000 ft, min insulation resistance after humidity exposure.
IDENTIFICATION OF PRODUCT: Required for sizes 22 and larger.
IDENTIFICATION, STRIPING, OR BANDING DURABILITY: 125 cycles (250 strokes); see Table II for test load.
IMPULSE DIELECTRIC TEST: 100% test; impulse voltage as specified in MIL-W-81381.
INSULATION RESISTANCE: 2500 megohms-1000 ft (min)
LAMINATION SEALING: Oven temperature, 230 ±2° C (446 ±3.6° F)
LIFE CYCLE: Oven temperature, 230 ±2° C (446 ±3.6° F) for 500 hours
MINIMUM WALL THICKNESS: Sizes 24 through 10: 8.0 mils
Sizes 8 through 2: 15.0 mils
POLYIMIDE CURE TEST: Applicable to sizes 10 and smaller
PROPELLANT RESISTANCE: Test not required
© RESIN COATING DURABILITY: Sizes 24 through 10: 250 cycles (500 strokes); see Table II for test load.
Sizes 8 through 2: Test not applicable.
SHRINKAGE: 0.031 inch (max) at 230 ±2° C (446 ±3.6° F)
SURFACE RESISTANCE: 5 megohms-inches (min), initial and final readings
THERMAL SHOCK: Oven temperature, 200 ±2° C (392 ±3.6° F) change in measurement, 0.031 inch (max)
WET DIELECTRIC TEST: 2500 volts (rms)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:21Z
Check the source to verify that this is the current version before use.

© WRINKLE TEST: Applicable to sizes 10 and smaller. No wrinkles shall be visible in the insulation at 3X magnification (3 diameters) after bending the wire one full turn around the mandrel specified in Table II. (The wire may be examined on the mandrel or after removal of the mandrel leaving the coil intact.) This test shall be included in the MIL-W-81381 quality conformance inspection as a Group II characteristic, one specimen to be tested from each sample unit.

Caution: This wire should not be subjected to physical contact with missile propellants.

Custodians:
Navy - AS
Army - EL
Air Force - 11

User activities:
Navy - MC, OS
Army - AT, AV, MU

Review activities:
Navy - EC, SH
Army - MI
Air Force - 99
DSA - IS
NSA

Preparing activity:
Navy - AS
(Project No. 6145-0705-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-w-81381/11C Wire, Electric, Fluorocarbon/Polyimide Insulated,...Etc.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Lakehurst, NJ 08733

POSTAGE AND FEES PAID
Navy Department
DOD-316

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department(Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:21Z
Check the source to verify that this is the current version before use.